

Demonstration of Proposed Features

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

Project: Credit Card Approval Prediction **Live URL:** <https://credit-card-approval-prediction-ra22.onrender.com>

Feature Demonstration Walkthrough

Feature 1: Model Comparison Dashboard

Steps:

1. Navigate to the home page (/)
2. Observe the model comparison table showing Logistic Regression, Decision Tree, Random Forest, and XGBoost
3. Random Forest is highlighted as the best model with 81.91% accuracy and 0.89 ROC-AUC
4. The hero section confirms "Best Model: Random Forest"

Expected Outcome: All 4 models with their metrics displayed in a clean table with the best model highlighted.

Feature 2: Scenario Pre-Fill Cards

Steps:

1. On the dashboard, click "Low Risk" scenario card
2. Browser navigates to /predict with all fields pre-filled with a low-risk profile
3. Click "Mid Risk" or "High Risk" to test different profiles

Expected Outcome: Prediction form is pre-filled with the selected scenario's values, ready for immediate submission.

Feature 3: Instant Prediction

Steps:

1. Navigate to /predict
2. Fill in applicant details (or use scenario pre-fill)
3. Click "Predict Approval"
4. Observe the result card appear with animation

Expected Outcome:

- Green card labeled "Approved" for low-risk applicant
 - Red card labeled "Rejected" for high-risk applicant
 - Confidence progress bar animates to the prediction probability
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Feature 4: Prediction History

Steps:

1. Make 3 predictions in sequence
2. Navigate to /history
3. Observe all 3 predictions listed with timestamps, result colors, and collapsible detail panels

Expected Outcome: Last 10 predictions displayed in reverse chronological order with summary statistics (X Approved / Y Rejected).

Feature 5: History Clear

Steps:

1. On the /history page, click "Clear History"
2. Page refreshes showing "No predictions yet" state

Expected Outcome: Session history is cleared; counter resets to 0.

Feature 6: Real-Time Income Display

Steps:

1. On the prediction form, type a value in the Annual Income field
2. Observe the label below the field update in real time

Expected Outcome: Income displays as formatted value (e.g., "150000.00 / year") as you type.

Feature 7: Responsive Layout

Steps:

1. Open the live URL on a mobile device or browser in responsive mode (375px width)
2. Check all pages: dashboard, predict form, history

Expected Outcome: All content fits within the viewport; navbar collapses to hamburger menu; form fields stack vertically.